

# INTERNATIONAL STANDARD

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**Coaxial communication cables –  
Part 5: Sectional specification for CATV trunk and distribution cables**



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**Coaxial communication cables –  
Part 5: Sectional specification for CATV trunk and distribution cables**

INTERNATIONAL  
ELECTROTECHNICAL  
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## INTERNATIONAL ELECTROTECHNICAL COMMISSION

## COAXIAL COMMUNICATION CABLES –

**Part 5: Sectional specification for CATV trunk  
and distribution cables**

## FOREWORD

- 1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as “IEC Publication(s)”). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
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International Standard IEC 61196-5 has been prepared by subcommittee 46A: Coaxial cables, of IEC technical committee 46: Cables, wires, waveguides, RF connectors, RF and microwave passive components and accessories.

This third edition cancels and replaces the second edition published in 2012. This edition constitutes a technical revision.

This edition includes the following significant technical change with respect to the previous edition:

- a) 1 Scope: revised, frequency range extended to 2 000 MHz;
- b) 4.3 Dielectric: life expectancy of the dielectric is proved by oxidative induction time (OIT) test before and after ageing according to IEC 60811-410;
- c) Annex A: Table A.1, Cable types added.

The text of this International Standard is based on the following documents:

| FDIS          | Report on voting |
|---------------|------------------|
| 46A/1351/FDIS | 46A/1357/RVD     |

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

This International Standard is to be used in conjunction with IEC 61196-1:2005.

A list of all parts of the IEC 61196 series, published under the general title *Coaxial communication cables*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

## COAXIAL COMMUNICATION CABLES –

### Part 5: Sectional specification for CATV trunk and distribution cables

#### 1 Scope

This part of IEC 61196, which is a sectional specification, applies to coaxial cables for analogue and digital one- and two-way signal transmission, e.g. for cable networks for television signals, sound signals and interactive services in accordance with IEC 60728-1, IEC 60728-1-1, IEC 60728-101, IEC 60728-10, ISO/IEC 11801-1 and ISO/IEC 11801-4. This includes also the transmission of BCT signals provided by a CATV, MATV or SMATV cable networks.

This document specifies the test procedures and requirements for trunk and distribution cables for temperatures between –40 °C and +65 °C and in the frequency range of 5 MHz to 2 000 MHz.

#### 2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60068-1:2013, *Environmental testing – Part 1: General and guidance*

IEC 60068-2-78, *Environmental testing – Part 2-78: Tests – Test Cab: Damp heat, steady state*

IEC 60096-0-1, *Radio Frequency cables – Part 0-1: Guide to the design of detail specifications – Coaxial cables*

IEC 60728-1, *Cable networks for television signals, sound signals and interactive services – Part 1: System performance of forward paths*

IEC 60811-410, *Electric and optical fibre cables – Test methods for non-metallic materials – Part 410: Miscellaneous tests – Test method for copper-catalyzed oxidative degradation of polyolefin insulated conductors*

IEC 60811-605, *Electric and optical fibre cables – Test methods for non-metallic materials Part 605: Physical tests – Measurement of carbon black and/or mineral filler in polyethylene compounds*

IEC 61196-1:2005, *Coaxial communication cables – Part 1: Generic specification – General, definitions and requirements*

IEC 61196-1-1, *Coaxial communication cables – Part 1-1: Capability approval for coaxial cables*

IEC 61196-1-101, *Coaxial communication cables – Part 1-101: Electrical test methods – Test for conductor d.c. resistance of cable*

IEC 61196-1-102, *Coaxial communication cables – Part 1-102: Electrical test methods – Test for insulation resistance of cable dielectric*

IEC 61196-1-105, *Coaxial communication cables – Part 1-105: Electrical test methods – Test for withstand voltage of cable dielectric*

IEC 61196-1-106, *Coaxial communication cables – Part 1-106: Electrical test methods – Test for withstand voltage of cable sheath*

IEC 61196-1-108, *Coaxial communication cables – Part 1-108: Electrical test methods – Test for characteristic impedance, phase and group delay, electrical length and propagation velocity*

IEC 61196-1-112, *Coaxial communication cables – Part 1-112: Electrical test methods – Test for return loss (uniformity of impedance)*

IEC 61196-1-113, *Coaxial communication cables – Part 1-113: Electrical test methods – Test for attenuation constant*

IEC 61196-1-115, *Coaxial communication cables – Part 1-115: Electrical test methods – Test for regularity of impedance (pulse/step function return loss)*

IEC 61196-1-201, *Coaxial communication cables – Part 1-201: Environmental test methods – Test for cold bend performance of cable*

IEC 61196-1-203, *Coaxial communication cables – Part 1-203: Environmental test methods – Test for water penetration of cable*

IEC 61196-1-206, *Coaxial communication cables – Part 1-206: Environmental test methods – Climatic sequence*

IEC 61196-1-209, *Coaxial communication cables – Part 1-209: Environmental test methods – Thermal ageing*

IEC 61196-1-301, *Coaxial communication cables – Part 1-301: Mechanical test methods – Test for ovality*

IEC 61196-1-302, *Coaxial communication cables – Part 1-302: Mechanical test methods – Test for eccentricity*

IEC 61196-1-308, *Coaxial communication cables – Part 1-308: Mechanical test methods – Test for tensile strength and elongation for copper-clad metals*

IEC 61196-1-310, *Coaxial communication cables – Part 1-310: Mechanical test methods – Test for torsion characteristics of copper-clad metals*

IEC 61196-1-313, *Coaxial communication cables – Part 1-313: Mechanical test methods – Adhesion of dielectric and sheath*

IEC 61196-1-314, *Coaxial communication cables – Part 1-314: Mechanical test methods – Test for bending*

IEC 61196-1-316, *Coaxial communication cables – Part 1-316: Mechanical test methods – Test of maximum pulling force of cable*



IEC 61196-1-317, *Coaxial communication cables – Part 1-317: Mechanical test methods – Test for crush resistance of cable*

IEC 61196-1-324, *Coaxial communication cables – Part 1-324: Mechanical test methods – Test for abrasion resistance of cable*

IEC 62153-1-1, *Metallic communication cables test methods – Part 1-1: Electrical – Measurement of the pulse/step return loss in the frequency domain using the Inverse Discrete Fourier Transformation (IDFT)*

IEC 62153-4-3, *Metallic communication cable test methods – Part 4-3: Electromagnetic compatibility (EMC) – Surface transfer impedance – Triaxial method*

IEC 62153-4-4, *Metallic communication cable test methods – Part 4-4: Electromagnetic compatibility (EMC) – Test method for measuring of the screening attenuation as up to and above 3 GHz*

EN 50289-4-17, *Communication cables – Specifications for test methods – Part 4-17: Test methods for UV resistance evaluation of the sheath of electrical and optical fibre cable*<sup>1</sup>

### 3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 61196-1 and IEC 60728-1 apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

### 4 Materials and cable construction

#### 4.1 Cable construction

The cable construction shall be in accordance with Subclauses 4.2 to 4.6 and the requirements stated in the relevant detail specification.

#### 4.2 Inner conductor

##### 4.2.1 Conductor material

Subclause 4.4.1 of 61196-1:2005 applies. The conductor material shall be as stated in the relevant cable detail specification.

##### 4.2.2 Conductor construction

The conductor shall consist of a single strand or tube.

In addition, 4.4.1 of IEC 61196-1:2005 applies.

The centre conductor diameter shall be stated in the relevant detail specification. The tolerance on the centre conductor shall be  $\pm 0,03$  mm for conductors with a diameter  $\leq 4$  mm.

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<sup>1</sup> An IEC test procedure for UV stability is under consideration.

For conductors with a diameter above 4 mm, the tolerance shall be stated in the detail specification.

### 4.3 Dielectric

The construction of the dielectric shall be one of the following:

- solid dielectric,
- air spaced dielectric,
- semi air spaced dielectric,
- gas-injected cellular polymer dielectric.

Life expectancy of the dielectric is proven by the oxidative induction time (OIT) test, before and after ageing, according to IEC 60811-410.

### 4.4 Outer conductor or screen

The type, material, nominal thickness and diameter of the outer conductor or screen shall be specified in the relevant detail specification. The tolerance of the outer conductor shall be  $\pm 0,05$  mm for constructions in accordance with 4.6.1 c) of IEC 61196-1:2005, excluding corrugated designs. The tolerance for all other constructions, including corrugated designs, shall be  $\pm 0,3$  mm in accordance with all other designs noted in 4.6.1 of IEC 61196-1:2005.

The construction and material of the outer conductor or screen shall be as stated in the relevant detail specification. The construction shall be in accordance with 4.6.1 c) or 4.6.1 f) or 4.6.1 g) of IEC 61196-1:2005.

For constructions with metal foil and/or braid, braid angle shall be between  $15^\circ$  and  $45^\circ$ . Coverage factor shall be specified in the detail specification.

### 4.5 Sheath

Subclause 4.7 of IEC 61196-1:2005 applies with the following amendments and additions.

Cables without an outer sheath shall not be subject to this subclause.

The outer sheath of the cable shall be a thermoplastic material as specified in the relevant detail specification.

The nominal sheath thickness shall be stated in the relevant detail specification.

The nominal diameter of the sheath shall be stated in the relevant detail specification.

The maximum allowable tolerance of the diameter shall be stated in the relevant detail specification.

The maximum allowable values for ovality and eccentricity are given in Table 4.

For aerial cables or cables for outdoor use with a black polyethylene sheath, the carbon black content shall be as specified in Table 4.

For other sheath material and colours of cables for outdoor use, the cable shall pass the UV stability test according to EN 50289-4-17.

The messenger type shall be specified in the relevant detail specification and shall include as a minimum the following criteria: type and material, tensile strength, corrosion properties and elongation.

#### **4.6 Completed cable**

The nominal dimensions and tolerances shall be stated in the detail specification. Preferred nominal dimensions and ratings are given in Annex A, Table A.1.

### **5 Standard ratings and characteristics**

The ratings and characteristics applicable to each cable shall be specified herein or in the relevant detail specification.

### **6 Identification and marking**

#### **6.1 Cable identification**

Subclause 6.1 of IEC 61196-1:2005 applies.

#### **6.2 Cable marking**

The cable marking shall be applied to the sheath or to the outer conductor when a sheath is not present. The marking shall consist of the IEC cable type number as given in 6.1.1 of IEC 61196-1:2005 and/or the manufacturer's designated markings when specified in the relevant cable specification.

#### **6.3 Labelling**

Labelling shall be provided in accordance with 6.3 of IEC 61196-1:2005 and the relevant detail specification.

### **7 Tests for completed cables**

#### **7.1 General**

When tested in accordance with the IEC 61196-1 series, the requirements given below shall apply.

Unless otherwise specified, all measurements shall be carried out under standard atmospheric conditions for testing in accordance with Clause 5 of IEC 60068-1:2013.

Applicable test methods shall be in accordance with the IEC 61196-1-100 series, IEC 61196-1-200 series, IEC 61196-1-300 series and the IEC 62153 series.

#### **7.2 Electrical testing of the finished cable**

Low-frequency and DC electrical test procedures and requirements are given in Table 1.

**Table 1 – Low-frequency and DC test procedures and requirements**

| No.       | IEC test procedure | Parameter                         | Requirements/remarks                                                                                                                                                                                                                                                                                      |
|-----------|--------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.1.1.1.1 | 61196-1-101        | Inner conductor resistance        | Applicable, value in accordance with the detail specification                                                                                                                                                                                                                                             |
| 7.1.1.1.2 | 61196-1-101        | Outer conductor resistance        | $\leq 15 \text{ m}\Omega/\text{m}$                                                                                                                                                                                                                                                                        |
| 7.1.1.2   | 61196-1-102        | Insulation resistance             | $\geq 104 \text{ M}\Omega \times \text{km}$                                                                                                                                                                                                                                                               |
| 7.1.1.3   | 61196-1-105        | Withstand voltage of dielectric   | 2 kV DC or 1,5 kV AC for 1 min, unless otherwise specified in the relevant detail specification                                                                                                                                                                                                           |
| 7.1.1.4   | 61196-1-106        | Withstand voltage of sheath (RMS) | Unless otherwise specified in the relevant detail specification,<br>Sheath thickness up to and including 0,5 mm = 1 kV<br>Sheath thickness over 0,5 mm and up to and including 0,8 mm = 2 kV<br>Sheath thickness over 0,8 mm and up to and including 1,0 mm = 3 kV<br>Sheath thickness over 1,0 mm = 5 kV |
| 7.1.1.5   | 60096-0-1          | Current carrying capacity         | Value in accordance with the detail specification                                                                                                                                                                                                                                                         |

### 7.3 High-frequency electrical and transmission test procedures and requirements

High-frequency electrical and transmission test procedures and requirements are given in Table 2.

**Table 2 – High-frequency electrical and transmission test procedures and requirements**

| No.     | IEC test procedure | Parameter                                      | Requirements/remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------|--------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.1.2.1 | 61196-1-108        | Characteristic impedance                       | $75 \Omega \pm 2 \Omega$ , unless the tolerance is otherwise specified in the relevant detail specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 7.1.2.2 | 61196-1-108        | Relative propagation velocity (velocity ratio) | May be specified for information purposes only in the detail specification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 7.1.2.3 | 61196-1-112        | Return loss                                    | RL $\geq 26 \text{ dB}$ from 5 MHz to 470 MHz,<br>RL $\geq 23 \text{ dB}$ from 470 MHz to 1 000 MHz<br>RL $\geq 20 \text{ dB}$ from 1 000 MHz to 2 000 MHz<br>The measurement inaccuracy $\Delta a_{r,f}$ shall be $< 1 \text{ dB}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 7.1.2.4 | 61196-1-113        | Attenuation constant                           | The maximum value at any frequency shall not be greater than calculated with the following formula:<br>$\alpha = 10 \times \lg(P_1/P_2) \times 100/l \text{ [dB/100 m]}$ .<br>where<br>$\alpha$ is the attenuation constant in dB/100 m (frequency dependent);<br>$P_1$ is the output power of a source where the load impedance and the source impedance are equal and of the same value as the nominal value of the specimen;<br>$P_2$ is the output power measured when the specimen is inserted into the test system, where the load impedance and the source impedance are equal and of the same value as the nominal value of the specimen.<br>$l$ is the physical length of the specimen in metres (m). |

| No.     | IEC test procedure | Parameter                                                     | Requirements/remarks                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------|--------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.1.2.5 | 61196-1-115        | Regularity of impedance                                       | Perform on both ends of tested cable<br>Regularity $\geq 40$ dB resp. $\leq 1$ %<br>Test procedure: IEC 61196-1-115 (time domain) or IEC 62153-1-1 (transformation from frequency domain into time domain by IDFT)                                                                                                                                                                                                                             |
| 7.1.2.6 | 62153-4-3          | Transfer impedance<br>After multiple bending:<br>see 7.3.9    | Class A: $\leq 5$ m $\Omega$ /m from 5 MHz to 30 MHz<br>Class A+: $\leq 2,5$ m $\Omega$ /m from 5 MHz to 30 MHz<br>Class A++: $\leq 0,9$ m $\Omega$ /m from 5 MHz to 30 MHz<br>Test procedure according to IEC 62153-4-3, triaxial method, after completion of the flexure test according to the relevant detail specification.                                                                                                                |
| 7.1.2.7 | 62153-4-4          | Screening attenuation<br>After multiple bending:<br>see 7.3.9 | Class A: $\geq 85$ dB from 30 MHz to 1 000 MHz<br>$\geq 75$ dB from 1 000 MHz to 2 000 MHz<br>Class A+: $\geq 95$ dB from 30 MHz to 1 000 MHz<br>$\geq 85$ dB from 1 000 MHz to 2 000 MHz<br>Class A++: $\geq 105$ dB from 30 MHz to 1 000 MHz<br>$\geq 95$ dB from 1 000 MHz to 2 000 MHz<br>Test procedure according to IEC 62153-4-4 (triaxial method) after completion of the flexure test according to the relevant detail specification. |

#### 7.4 Environmental test procedures and requirements of the finished cable

Environmental test procedures and requirements are given in Table 3.

**Table 3 – Environmental test procedures and requirements of the finished cable**

| No.   | IEC test procedure | Parameter                | Requirements/remarks                                                                                                                                                                                                                                   |
|-------|--------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.2.1 | 61196-1-201        | Cold bend performance    | Test method A, B, or C as specified in the relevant detail specification.<br>The test temperature shall be stated in the relevant detail specification.<br>No physical damages of conductors, dielectric and sheath.                                   |
| 7.2.2 | 61196-1-203        | Water penetration        | When required, in accordance with the relevant detail specification.                                                                                                                                                                                   |
| 7.2.3 | 61196-1-206        | Climatic sequence        | $T_A = -40$ °C; $T_B = +65$ °C; $t_1 = 24$ h, unless otherwise specified in the detail specification.<br>Number of cycles: 3<br>Influenced mechanical and electrical characteristics shall be tested and defined in the relevant detail specification. |
| 7.2.4 | 60068-2-78         | Damp heat (steady state) | Influenced mechanical and electrical characteristics shall be tested and defined in the relevant detail specification.<br>1) Test temperature<br>2) Percentage relative humidity<br>3) Test duration                                                   |

| No.                                                                                                                                                               | IEC test procedure         | Parameter                                     | Requirements/remarks                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.2.5                                                                                                                                                             | EN 50289-4-17 <sup>a</sup> | Ultraviolet stability of the sheath or jacket | All cables for outdoor or other applications that are subjected to UV radiation shall meet the following requirements.<br>No visual cracks<br>– change in elongation $\pm 20$ % max. after 720 h<br>– change in tensile strength $\pm 20$ % max. after 720 h |
| 7.2.6                                                                                                                                                             | 61196-1-209                | Thermal ageing                                | Transmission characteristics shall remain within the specified limits as defined in the relevant detail specification                                                                                                                                        |
| 7.2.7                                                                                                                                                             | 60811-410                  | Oxidative Induction time (OIT)                | Requirements for OIT – Initial: 20 min minimum, after aging: 70 % of initial value                                                                                                                                                                           |
| 7.2.8                                                                                                                                                             | 60811-605                  | Carbon black content                          | Only for cables with black PE; $\geq 2,0$ % <sup>a</sup>                                                                                                                                                                                                     |
| <sup>a</sup> In some regions, a value of carbon black content of (2,5 % $\pm$ 0,5 %) of the sheath is required. In this case, UV stability test is not necessary. |                            |                                               |                                                                                                                                                                                                                                                              |

## 7.5 Test procedures and requirements of mechanical characteristics of the finished cable

Test procedures and requirements for mechanical characteristics of the finished cable are given in Table 4.

**Table 4 – Test procedures and requirements of mechanical characteristics of the finished cable**

| No.   | IEC test procedure | Parameter                                                                    | Requirements/remarks                                                                                                                                                                                                                                                                |
|-------|--------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.3.1 | 61196-1-301        | Ovality of outer conductor or screen                                         | $\leq 7$ %                                                                                                                                                                                                                                                                          |
| 7.3.2 | 61196-1-301        | Ovality of the sheath                                                        | $\leq 7$ %                                                                                                                                                                                                                                                                          |
| 7.3.3 | 61196-1-302        | Eccentricity of dielectric                                                   | $\leq 10$ %                                                                                                                                                                                                                                                                         |
| 7.3.4 | 61196-1-302        | Eccentricity of the sheath                                                   | $\leq 10$ %                                                                                                                                                                                                                                                                         |
| 7.3.5 | 61196-1-308        | Tensile strength and elongation of the copper or copper-clad inner conductor | Shall be in accordance with 4.4.1 of IEC 61196-1:2005                                                                                                                                                                                                                               |
| 7.3.6 | 61196-1-310        | Torsion test for copper-clad metals                                          | The surface shall not reveal any irregular seams, pits or slivers of sufficient magnitude or inherent defects.                                                                                                                                                                      |
| 7.3.7 | 61196-1-313        | Adhesion testing                                                             | Inner conductor to dielectric, sample length = 50 mm<br>Pressure force $F_a$ required to remove dielectric shall be $0,1 \text{ MPa} \leq F_a \leq 1,0 \text{ MPa}$ . <sup>a, b</sup>                                                                                               |
| 7.3.8 | 61196-1-314        | Bending characteristics                                                      | Single bending: According to the detail specification<br>Multiple bending: Test method to be performed per 8.3.2, procedure 1<br>Radius of mandrel: as specified by the manufacturer<br>Tension: as defined in row 7.3.10 of Table 4.<br>Speed: $\leq 1$ m/s<br>Number of cycles: 3 |
| 7.3.9 | 61196-1-316        | Tensile strength of cable (longitudinal pull)                                | According to the detail specification                                                                                                                                                                                                                                               |

| No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | IEC test procedure | Parameter                 | Requirements/remarks                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.3.10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 61196-1-317        | Crush resistance of cable | Load = According to the detail specification, applied for 2 min<br><br>After a 2 min recovery time, the maximum impedance irregularity shall be $\leq 1\%$ , when measured in accordance with IEC 61196-1-115.<br><br>No physical damage of the sheath or jacket |
| 7.3.11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 61196-1-324        | Abrasion resistance       | According to the detail specification                                                                                                                                                                                                                            |
| <p><sup>a</sup> The adhesion of the dielectric to the inner conductor, <math>F_a</math>, is given in MPa by the following formula:</p> $F_a = \frac{F}{\pi \cdot d \cdot l}.$ <p>where <math>F</math> is the force, <math>d</math> is the diameter of inner conductor, <math>l</math> is the length of the sample.</p> <p><sup>b</sup> Other values may be required if special tools for preparing connector mounting are used (see relevant detail specification). Limits should be given in the detail specification.</p> |                    |                           |                                                                                                                                                                                                                                                                  |

## 7.6 Fire performance

When intended to be installed in buildings, these cables may fall under the requirements of local, regional or governmental fire and safety standards as outlined in Table 5.

**Table 5 – Fire performance requirements**

| No.   | Test method/standard | Parameter                   | Requirements/remarks – according to the relevant detail specification |
|-------|----------------------|-----------------------------|-----------------------------------------------------------------------|
| 7.4.1 |                      | Flame propagation           |                                                                       |
| 7.4.2 |                      | Acid gas emission           |                                                                       |
| 7.4.3 |                      | Smoke generation            |                                                                       |
| 7.4.4 |                      | Toxic gas emission          |                                                                       |
| 7.4.5 |                      | To be defined (as required) |                                                                       |

## 8 Quality assessment

When specified in the sectional or detail specifications, quality procedures shall be in accordance with IEC 61196-1-1.

## 9 Delivery and storage

Delivery of cables shall be in accordance with Clause 9 of IEC 61196-1:2005.

## Annex A (informative)

### Cable types

Table A.1 indicates typical cables and properties for informative purposes with copper conductors.

Alternative conductor materials, dimensions and characteristics may be defined in the detail specification.

**Table A.1 – Distribution and trunk cables – Preferred nominal dimensions and ratings**

| Type <sup>a</sup>                                                                                                                                                                                                                                                                                                                                                                                                                | 13<br>A/A+/A++ | 10<br>A/A+/A++ | 9<br>A/A+/A++ | 6,1<br>A/A+/A++ | 3<br>A/A+/A++ |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------|---------------|-----------------|---------------|
| <b>Characteristic</b>                                                                                                                                                                                                                                                                                                                                                                                                            |                |                |               |                 |               |
| <b>Nominal diameter [mm]</b>                                                                                                                                                                                                                                                                                                                                                                                                     |                |                |               |                 |               |
| over dielectric                                                                                                                                                                                                                                                                                                                                                                                                                  | 7,2            | 6,9            | 9             | 13,5            | 20            |
| outer diameter                                                                                                                                                                                                                                                                                                                                                                                                                   | 10,5           | 10,5           | 12,5          | 17,5            | 25            |
| <b>Attenuation max.</b><br>[dB/100 m]                                                                                                                                                                                                                                                                                                                                                                                            |                |                |               |                 |               |
| at 200 MHz                                                                                                                                                                                                                                                                                                                                                                                                                       | 6              | 4,8            | 4,5           | 3               | 2             |
| at 800 MHz                                                                                                                                                                                                                                                                                                                                                                                                                       | 13             | 10             | 9             | 6               | 4             |
| at 2 400 MHz                                                                                                                                                                                                                                                                                                                                                                                                                     | 24             | 17             | 16            | 11              | 7,4           |
| <b>Attenuation coefficient <sup>b</sup></b>                                                                                                                                                                                                                                                                                                                                                                                      |                |                |               |                 |               |
| a                                                                                                                                                                                                                                                                                                                                                                                                                                | 0,39           | 0,31           | 0,29          | 0,18            | 0,12          |
| b                                                                                                                                                                                                                                                                                                                                                                                                                                | 0,001 8        | 0,000 7        | 0,000 7       | 0,000 7         | 0,000 6       |
| c                                                                                                                                                                                                                                                                                                                                                                                                                                | 0,25           | 0,15           | 0,15          | 0,10            | 0,10          |
| <b>Screening class</b>                                                                                                                                                                                                                                                                                                                                                                                                           | A/A+/A++       | A/A+/A++       | A/A+/A++      | A/A+/A++        | A/A+/A++      |
| <b>Max. DC current <sup>c</sup> [A]</b>                                                                                                                                                                                                                                                                                                                                                                                          | 13             | 14,5           | 20            | 30              | 66            |
| <sup>a</sup> Type designation is given by: IEC Standard No., attenuation at 800 MHz (rounded to an integer) and screening class, e.g. 61196-5-1 – 9 – A+.                                                                                                                                                                                                                                                                        |                |                |               |                 |               |
| <sup>b</sup> Attenuation vs. frequency: $a(f) = a \cdot \sqrt{f} + b \cdot f + c$ . [dB/100 m] for copper conductors. In the case of copper clad conductors, a term $d/\sqrt{f}$ shall be added: $a \cdot \sqrt{f} + b \cdot f + c + d/\sqrt{f}$ , $d$ according to the relevant detail specification. Coefficients $a$ , $b$ , $c$ and $d$ can be determined from the fitted attenuation curve, using the least squares method. |                |                |               |                 |               |
| <sup>c</sup> Calculated value for free air installation at room temperature.                                                                                                                                                                                                                                                                                                                                                     |                |                |               |                 |               |





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